Azure 101: Core Services Beginner Cheat Sheet

Azure provides hundreds of services, but you only need a handful to build and deploy modern cloud applications. Mastering these fundamental building blocks gives you the foundation to deploy apps, manage costs, and secure workloads with confidence.

What you'll learn:

1. Resource Groups: Organizing Your Cloud Assets
2. Azure Virtual Machines: On-Demand Compute Power
3. Azure Blob Storage: Low-Cost Scalable File Storage
4. Azure Key Vault: Protecting Secrets and Credentials

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Resource Groups: Organizing Your Cloud Assets

A Resource Group is a logical container for resources deployed on Azure, like databases, web apps, and virtual networks. Think of it like a project folder on your computer desktop. Grouping resources together makes it easy to track project costs, grant team access, and delete everything at once when testing ends. Every resource must belong to exactly one Resource Group.

```
$az group create --name demo-project-rg --location eastus
```

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Azure Virtual Machines: On-Demand Compute Power

Azure Virtual Machines provide Infrastructure as a Service (IaaS) by renting you a virtual server running Linux or Windows. You choose the CPU cores, memory size, and storage disks that your application requires. They are perfect when you need low-level operating system control or want to run custom system daemons. Always ensure you add SSH keys during creation so you can log in securely.

```
$az vm create --resource-group demo-project-rg --name my-linux-vm --image Ubuntu2204 --admin-username
```

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Azure Blob Storage: Low-Cost Scalable File Storage

Blob Storage lets you store massive amounts of unstructured data like images, audio, video files, and system logs. You first create a Storage Account, then create a Container inside it, and finally upload your files as blobs. It automatically handles replication and high availability without any disk provisioning. You can access stored files through standard HTTP/HTTPS endpoints anywhere.

```
$az storage blob upload --account-name mystorageacct --container-name media --file ./image.png --name
```

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Azure Key Vault: Protecting Secrets and Credentials

Azure Key Vault solves the dangerous problem of committing passwords and API keys into source code repositories. It acts as a centralized cloud locker where you store secrets, SSL certificates, and encryption keys with strict role-based access. Applications query Key Vault dynamically at startup to load secrets directly into memory. This ensures credentials stay secret even if your repository becomes public.

```
$az keyvault secret set --vault-name my-secure-vault --name db-connection-string --value "Server=tcp:
```

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Key Takeaways

+ Keep all resources for a single application inside one Resource Group for easy cleanup.
+ Set spending alerts and budget caps on your Azure subscription to prevent unexpected bills.
+ Never put sensitive API keys or connection strings in Git; always fetch them from Azure Key Vault.
+ Stop and deallocate unused Virtual Machines during nights or weekends to stop compute charges.
+ Always lock your resources with Read-Only or Cannot-Delete management locks in production.

Watch Out For

! Leaving virtual machines running after lab practice and draining subscription credits.
! Opening port 22 (SSH) or port 3389 (RDP) to 0.0.0.0/0 instead of your own IP address.
! Storing persistent app data on the temporary VM drive (D: on Windows, /dev/sdb1 on Linux) which wipes on reboot.
! Forgetting to use lowercase letters and numbers only when naming Azure Storage Accounts.

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